

Jacob Lindland

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EDUCATION

University of Florida, Herbert Wertheim College of Engineering

May 2027

Bachelor of Science in Mechanical Engineering | Minor in Computer Science

6 x Dean's List | GPA: 3.75/4.0

- Courses: Heat Transfer, Fluids, Mechanical Design 1&2, Vibrations, Design and Manufacturing Lab, Numerical Analysis

WORK EXPERIENCE

GE Aerospace

May 2026 – Present

Engines Engineering Intern, Advanced Measurement Technologies

Evendale, OH

- Performed thermal simulation and analysis in ANSYS to validate design of IR camera probe for deployment in RISE compact core High Pressure Turbine nozzle to enable safe instrumentation of hot gas path region
- Mitigated potential assembly failures by correlating stack-up tolerances with thermal growth projections, successfully validating critical thermal loading parameters.
- Supported ongoing root cause investigation by manufacturing thin-film sensor arrays in lab environment using RF sputtering and photolithography, collecting and processing relevant micron-thickness and insulation resistance data
- Managed specification and assembly of custom hardware for GE Research H2 combustor test rig, reducing order costs by \$1000+

Mechanical and Aerospace Engineering, University of Florida

Jan. 2026 – Present

Teaching Assistant, Numerical Methods of Analysis

Gainesville, FL

- Hosted weekly office hours for 300+ students to review lecture content, homework issues, and break down complex topics
- Graded and evaluated 200+ implementations of numerical methods across homework and exams, providing feedback

GE Appliances, a Haier Company

Sep. 2024 – Dec. 2024

Manufacturing Engineering Co-op, Manufacturing & Supplier Quality

Louisville, KY

- Collaborated on lean team to design and prototype alignment fixture to eliminate 80% downtime caused by common line stoppage
- Developed analytic to track 10,000+ granular Electrostatic Dissipative safety compliance indicators, enhancing problem visibility and enabling a 90% boost in ESD compliance over 3 months
- Eliminated 300+ hrs/year of work by automating a compliance tracking and reporting system with scripting and data visualization

PROJECTS & PUBLICATIONS

Autonomous Bamboo Harvester Robot

Aug. 2025 – Present

Undergraduate Researcher under Dr. Medeiros, IFAS

- Engineered custom electric pruner end-effector mount for 6-axis robot arm with appropriate bearings, fasteners, GD&T, gear train
- Defined loading & vibrational conditions of various cutting configurations to design mechanical support for sawblade
- Developed ROS2-python reinforcement learning pipeline to train realtime YOLO model to identify and segment bamboo culms

Mars Rover Sky Crane Control System

May. 2026

Undergraduate Project

- Designed and implemented PID and state-feedback controllers for a rover system in Simulink, using frequency-domain loop shaping on Bode plots to optimize phase/gain margins and ensure stability against non-linear dynamics
- Validated system performance and mitigated noise by implementing a high-fidelity simulation model for the multi-stage landing system using MATLAB state-space matrices and non-linear Simulink dynamics architecture

Adapting Lunar Regolith for In-situ Resource Utilization

May. 2025 – Aug. 2025

Undergraduate Researcher under Dr. Ruzyski, MSE

- Developed novel Lunar regolith PLA matrix composites suitable for FDM printing and manufacturing for lunar habitat construction
- Characterized Regolith-PLA composite material by conducting Instron UTM uniaxial loading tests and applying statistical analysis

INVOLVEMENT

Theta Tau Professional Engineering Society

Sep. 2023 – Present

President

Gainesville, FL

- Lead weekly chapters to 110+ members, maintaining parliamentary procedure and assessing & directing overall chapter progress
- Directed executive board and managed \$150,000+ treasury, improving and standardizing financial transparency initiatives
- Represented all members to national organization, responsible for all safety and risk operations under national oversight

Vice President

- Oversaw 25+ functional committees to drive recruitment and member professional development, resulting in the most successful recruitment cycle in recent chapter history and record-attendance private career showcase
- Leveraged long-term event planning to execute the chapters vision for the semester full of fulfilling events and opportunities

SKILLS

- Presentational communication, ANSYS Workbench/Mechanical, stack-up analysis, fastener selection, statistical analysis, rapid prototyping (FDM, water jet, laser cutter), vendor selection, DFM, statistical analysis, MATLAB, Simulink, Solidworks simulation